

SOM 787 Spatial Analytics-I Class Assignment 01

Objective:Analysing changes of Prevalence of crimes against women across districts and states in India from 2020 to 2022

Install and Load packages

Install all the necessary libraries

```
# Define a vector of required packages
packages <- c(
  "tidyverse",      # For data manipulation and visualization
  "sf",             # For simple feature operations (handling spatial data)
  "here",           # For managing file paths
  "patchwork",      # For organizing multiple plots
  "raster",          # For handling raster data
  "exactextractr",  # For extracting data from raster objects
  "tidygeocoder",   # For geocoding (converting addresses to geographic coordinates)
  "ggmap",           # For spatial visualization and mapping
  "ggspatial",      # For adding spatial context to ggplot2 visualizations
  "epitrix",         # For epidemiological analysis
  "janitor",         # For data cleaning
  "leaflet",         # For interactive maps
  "HistData",        # For historical datasets
  "gtsummary",       # For creating summary tables
  "spdep",           # For Spatial Dependence: Weighting Schemes, Statistics
  "tmap",            # For thematic maps
  "stringdist",      # For fuzzy text search and approximate string matching
  "viridis",         # For viridis palette for discrete data
  "wesanderson"     # For Inspired by Wes Anderson's movies, offering aesthetic, subtle palette
)
```

```
# Extract not installed packages
not_installed <- packages[!(packages %in% installed.packages()[ , "Package"])]

# Install not installed packages
if(length(not_installed)) install.packages(not_installed)
```

Load packages

```
suppressPackageStartupMessages(lapply(packages, library, character.only = TRUE))
```

```
## [[1]]
## [1] "lubridate" "forcats"   "stringr"   "dplyr"     "purrr"     "readr"
## [7] "tidyr"     "tibble"    "ggplot2"   "tidyverse" "stats"     "graphics"
## [13] "grDevices" "utils"     "datasets"  "methods"   "base"
##
## [[2]]
## [1] "sf"         "lubridate" "forcats"   "stringr"   "dplyr"     "purrr"
## [7] "readr"      "tidyr"     "tibble"    "ggplot2"   "tidyverse" "stats"
## [13] "graphics"  "grDevices" "utils"     "datasets"  "methods"   "base"
```

```

##
## [[3]]
## [1] "here"      "sf"      "lubridate" "forcats" "stringr" "dplyr"
## [7] "purrr"    "readr"   "tidyr"    "tibble"  "ggplot2" "tidyverse"
## [13] "stats"    "graphics" "grDevices" "utils"   "datasets" "methods"
## [19] "base"
##
## [[4]]
## [1] "patchwork" "here"      "sf"      "lubridate" "forcats" "stringr"
## [7] "dplyr"     "purrr"    "readr"   "tidyr"    "tibble"  "ggplot2"
## [13] "tidyverse" "stats"    "graphics" "grDevices" "utils"   "datasets"
## [19] "methods"   "base"
##
## [[5]]
## [1] "raster"    "sp"      "patchwork" "here"      "sf"      "lubridate"
## [7] "forcats"   "stringr" "dplyr"     "purrr"    "readr"   "tidyr"
## [13] "tibble"    "ggplot2" "tidyverse" "stats"    "graphics" "grDevices"
## [19] "utils"     "datasets" "methods"   "base"
##
## [[6]]
## [1] "exactextractr" "raster"      "sp"      "patchwork"
## [5] "here"          "sf"          "lubridate" "forcats"
## [9] "stringr"       "dplyr"       "purrr"    "readr"
## [13] "tidyr"         "tibble"      "ggplot2"  "tidyverse"
## [17] "stats"         "graphics"    "grDevices" "utils"
## [21] "datasets"      "methods"     "base"
##
## [[7]]
## [1] "tidygeocoder" "exactextractr" "raster"      "sp"
## [5] "patchwork"     "here"          "sf"          "lubridate"
## [9] "forcats"       "stringr"       "dplyr"       "purrr"
## [13] "readr"         "tidyr"         "tibble"      "ggplot2"
## [17] "tidyverse"     "stats"         "graphics"    "grDevices"
## [21] "utils"         "datasets"      "methods"     "base"
##
## [[8]]
## [1] "ggmap"      "tidygeocoder" "exactextractr" "raster"
## [5] "sp"         "patchwork"    "here"          "sf"
## [9] "lubridate"  "forcats"      "stringr"       "dplyr"
## [13] "purrr"     "readr"        "tidyr"         "tibble"
## [17] "ggplot2"   "tidyverse"    "stats"         "graphics"
## [21] "grDevices" "utils"        "datasets"      "methods"
## [25] "base"
##
## [[9]]
## [1] "ggspatial" "ggmap"      "tidygeocoder" "exactextractr"
## [5] "raster"     "sp"         "patchwork"    "here"
## [9] "sf"         "lubridate"  "forcats"      "stringr"
## [13] "dplyr"      "purrr"     "readr"        "tidyr"
## [17] "tibble"     "ggplot2"   "tidyverse"    "stats"
## [21] "graphics"   "grDevices" "utils"        "datasets"
## [25] "methods"    "base"
##
## [[10]]

```

```

## [1] "epitrix"      "ggspatial"    "ggmap"        "tidygeocoder"
## [5] "exactextractr" "raster"       "sp"           "patchwork"
## [9] "here"         "sf"           "lubridate"    "forcats"
## [13] "stringr"      "dplyr"        "purrr"        "readr"
## [17] "tidyr"        "tibble"       "ggplot2"      "tidyverse"
## [21] "stats"        "graphics"     "grDevices"    "utils"
## [25] "datasets"     "methods"     "base"
##
## [[11]]
## [1] "janitor"      "epitrix"      "ggspatial"    "ggmap"
## [5] "tidygeocoder" "exactextractr" "raster"       "sp"
## [9] "patchwork"    "here"         "sf"           "lubridate"
## [13] "forcats"      "stringr"      "dplyr"        "purrr"
## [17] "readr"        "tidyr"        "tibble"       "ggplot2"
## [21] "tidyverse"    "stats"        "graphics"     "grDevices"
## [25] "utils"        "datasets"     "methods"     "base"
##
## [[12]]
## [1] "leaflet"      "janitor"      "epitrix"      "ggspatial"
## [5] "ggmap"        "tidygeocoder" "exactextractr" "raster"
## [9] "sp"           "patchwork"    "here"         "sf"
## [13] "lubridate"    "forcats"      "stringr"      "dplyr"
## [17] "purrr"        "readr"        "tidyr"        "tibble"
## [21] "ggplot2"      "tidyverse"    "stats"        "graphics"
## [25] "grDevices"    "utils"        "datasets"     "methods"
## [29] "base"
##
## [[13]]
## [1] "HistData"     "leaflet"      "janitor"      "epitrix"
## [5] "ggspatial"    "ggmap"        "tidygeocoder" "exactextractr"
## [9] "raster"       "sp"           "patchwork"    "here"
## [13] "sf"           "lubridate"    "forcats"      "stringr"
## [17] "dplyr"        "purrr"        "readr"        "tidyr"
## [21] "tibble"       "ggplot2"      "tidyverse"    "stats"
## [25] "graphics"     "grDevices"    "utils"        "datasets"
## [29] "methods"     "base"
##
## [[14]]
## [1] "gtsummary"    "HistData"     "leaflet"      "janitor"
## [5] "epitrix"      "ggspatial"    "ggmap"        "tidygeocoder"
## [9] "exactextractr" "raster"       "sp"           "patchwork"
## [13] "here"         "sf"           "lubridate"    "forcats"
## [17] "stringr"      "dplyr"        "purrr"        "readr"
## [21] "tidyr"        "tibble"       "ggplot2"      "tidyverse"
## [25] "stats"        "graphics"     "grDevices"    "utils"
## [29] "datasets"     "methods"     "base"
##
## [[15]]
## [1] "spdep"        "spData"       "gtsummary"    "HistData"
## [5] "leaflet"      "janitor"      "epitrix"      "ggspatial"
## [9] "ggmap"        "tidygeocoder" "exactextractr" "raster"
## [13] "sp"           "patchwork"    "here"         "sf"
## [17] "lubridate"    "forcats"      "stringr"      "dplyr"
## [21] "purrr"        "readr"        "tidyr"        "tibble"

```

```

## [25] "ggplot2"      "tidyverse"    "stats"        "graphics"
## [29] "grDevices"   "utils"        "datasets"     "methods"
## [33] "base"
##
## [[16]]
## [1] "tmap"      "spdep"      "spData"      "gtsummary"
## [5] "HistData"  "leaflet"    "janitor"     "epitrix"
## [9] "ggspatial" "ggmap"      "tidygeocoder" "exactextractr"
## [13] "raster"    "sp"         "patchwork"   "here"
## [17] "sf"        "lubridate"  "forcats"     "stringr"
## [21] "dplyr"     "purrr"     "readr"       "tidyr"
## [25] "tibble"    "ggplot2"    "tidyverse"   "stats"
## [29] "graphics"  "grDevices"  "utils"       "datasets"
## [33] "methods"   "base"
##
## [[17]]
## [1] "stringdist"  "tmap"      "spdep"      "spData"
## [5] "gtsummary"   "HistData"  "leaflet"    "janitor"
## [9] "epitrix"     "ggspatial" "ggmap"      "tidygeocoder"
## [13] "exactextractr" "raster"    "sp"         "patchwork"
## [17] "here"        "sf"        "lubridate"  "forcats"
## [21] "stringr"     "dplyr"     "purrr"     "readr"
## [25] "tidyr"       "tibble"    "ggplot2"    "tidyverse"
## [29] "stats"       "graphics"  "grDevices"  "utils"
## [33] "datasets"    "methods"   "base"
##
## [[18]]
## [1] "viridis"      "viridisLite" "stringdist"  "tmap"
## [5] "spdep"        "spData"      "gtsummary"   "HistData"
## [9] "leaflet"     "janitor"     "epitrix"     "ggspatial"
## [13] "ggmap"       "tidygeocoder" "exactextractr" "raster"
## [17] "sp"          "patchwork"   "here"        "sf"
## [21] "lubridate"    "forcats"     "stringr"     "dplyr"
## [25] "purrr"       "readr"       "tidyr"       "tibble"
## [29] "ggplot2"     "tidyverse"   "stats"       "graphics"
## [33] "grDevices"   "utils"       "datasets"    "methods"
## [37] "base"
##
## [[19]]
## [1] "wesanderson" "viridis"     "viridisLite" "stringdist"
## [5] "tmap"         "spdep"      "spData"      "gtsummary"
## [9] "HistData"     "leaflet"    "janitor"     "epitrix"
## [13] "ggspatial"    "ggmap"      "tidygeocoder" "exactextractr"
## [17] "raster"       "sp"         "patchwork"   "here"
## [21] "sf"          "lubridate"  "forcats"     "stringr"
## [25] "dplyr"       "purrr"     "readr"       "tidyr"
## [29] "tibble"     "ggplot2"    "tidyverse"   "stats"
## [33] "graphics"    "grDevices"  "utils"       "datasets"
## [37] "methods"     "base"

```

Load and check/clean Data

Load Non-Spatial Data


```
# Load the crime data
crime_df <- read_csv(here("data", "crime_dateset_india - Sheet1.csv"))

## Rows: 706 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (2): DISTRICT, state name
## dbl (8): Sl. No, censuscode, RR_2020, LL_2020, UL_2020, RR_2022, LL_2022, UL...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Drop NA rows
crime_df <- crime_df |> drop_na()
```

```
# Lowercase all data entries
crime_df <- data.frame(lapply(crime_df, function(x) {
  if (is.character(x)) {
    return(tolower(x))
  } else {
    return(x)
  }
}), stringsAsFactors = FALSE)
```

```
# Explore the crime data
crime_df |>
  glimpse()
```

```
## Rows: 635
## Columns: 10
## $ Sl.No      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, ~
## $ DISTRICT   <chr> "kupwara", "badgam", "leh (ladakh)", "kargil", "punch", "ra~
## $ state.name <chr> "jammu & kashmir", "jammu & kashmir", "jammu & kashmir", "j~
## $ censuscode <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, ~
## $ RR_2020    <dbl> 1.13, 1.52, 0.20, 0.20, 0.90, 1.00, 0.69, 1.32, 0.62, 0.81,~
## $ LL_2020    <dbl> 1.00, 1.36, 0.09, 0.11, 0.75, 0.86, 0.58, 1.19, 0.48, 0.72,~
## $ UL_2020    <dbl> 1.26, 1.68, 0.37, 0.33, 1.06, 1.15, 0.82, 1.45, 0.76, 0.91,~
## $ RR_2022    <dbl> 0.78, 1.06, 0.23, 0.25, 0.81, 1.40, 0.44, 0.61, 1.33, 1.07,~
## $ LL_2022    <dbl> 0.69, 0.95, 0.12, 0.14, 0.68, 1.25, 0.36, 0.53, 1.15, 0.97,~
## $ UL_2022    <dbl> 0.89, 1.19, 0.40, 0.39, 0.95, 1.56, 0.53, 0.69, 1.54, 1.17,~
```

```
# Clean the column names using the 'clean_names' function from the 'janitor' package and display the new names
# This standardizes column names to lower_case_with_underscores
# Rename columns
```

```
colnames(crime_df) <- crime_df |> janitor::clean_names() |> names()
```

```
head(crime_df)
```

```
##   sl_no    district    state_name censuscode rr_2020 ll_2020 ul_2020 rr_2022
## 1     1      kupwara jammu & kashmir         1    1.13    1.00    1.26    0.78
## 2     2      badgam jammu & kashmir         2    1.52    1.36    1.68    1.06
## 3     3 leh (ladakh) jammu & kashmir         3    0.20    0.09    0.37    0.23
## 4     4      kargil jammu & kashmir         4    0.20    0.11    0.33    0.25
## 5     5      punch jammu & kashmir         5    0.90    0.75    1.06    0.81
## 6     6    rajouri jammu & kashmir         6    1.00    0.86    1.15    1.40
##   ll_2022 ul_2022
```

```
## 1    0.69    0.89
## 2    0.95    1.19
## 3    0.12    0.40
## 4    0.14    0.39
## 5    0.68    0.95
## 6    1.25    1.56
```

Load Spatial Data

```
# Load the India district shape file
india_district_sf <- read_rds(here("spatial_files", "india_district_sf.rds"))

# Rename columns
colnames(india_district_sf) <- c("district", "geometry")

# Explore the India district shape file
india_district_sf |>
  glimpse()

## Rows: 726
## Columns: 2
## $ district <chr> "adilabad_telangana", "agar_malwa_madhya_pradesh", "agra_uttar-
## $ geometry <MULTIPOLYGON [°]> MULTIPOLYGON (((78.84098 19..., MULTIPOLYGON (((~

# Convert the spatial data into Geodetic CRS
india_district_sf <- india_district_sf |>
  st_transform(4326)

# Check the CRS of the spatial data
india_district_sf |> st_crs()

## Coordinate Reference System:
##   User input: EPSG:4326
##   wkt:
##   GEOGCRS["WGS 84",
##     ENSEMBLE["World Geodetic System 1984 ensemble",
##       MEMBER["World Geodetic System 1984 (Transit)"],
##       MEMBER["World Geodetic System 1984 (G730)"],
##       MEMBER["World Geodetic System 1984 (G873)"],
##       MEMBER["World Geodetic System 1984 (G1150)"],
##       MEMBER["World Geodetic System 1984 (G1674)"],
##       MEMBER["World Geodetic System 1984 (G1762)"],
##       MEMBER["World Geodetic System 1984 (G2139)"],
##     ELLIPSOID["WGS 84",6378137,298.257223563,
##       LENGTHUNIT["metre",1]],
##     ENSEMBLEACCURACY[2.0]],
##     PRIMEM["Greenwich",0,
##       ANGLEUNIT["degree",0.0174532925199433]],
##     CS[ellipsoidal,2],
##       AXIS["geodetic latitude (Lat)",north,
##         ORDER[1],
##         ANGLEUNIT["degree",0.0174532925199433]],
##       AXIS["geodetic longitude (Lon)",east,
##         ORDER[2],
##         ANGLEUNIT["degree",0.0174532925199433]],
```

```

##      USAGE[
##          SCOPE["Horizontal component of 3D system."],
##          AREA["World."],
##          BBOX[-90,-180,90,180]],
##      ID["EPSG",4326]]

# To clean the state district column of india_district_sf -
# Cleaning the India district shape file
# Define a list of state and union territories names

state_union_names <- c(
  # States
  "andhra_pradesh", "arunachal_pradesh", "assam", "bihar", "chhattisgarh", "goa",
  "gujarat", "haryana", "himachal_pradesh", "jharkhand", "karnataka", "kerala",
  "madhya_pradesh", "maharashtra", "manipur", "meghalaya", "mizoram", "nagaland",
  "odisha", "punjab", "rajasthan", "sikkim", "tamil_nadu", "telangana",
  "tripura", "uttar_pradesh", "uttarakhand", "west_bengal",

  # Union Territories
  "andaman_nicobar_islands", "chandigarh", "dadra_nagar_haveli", "daman_diu",
  "lakshadweep", "nct_delhi", "puducherry", "ladakh", "jammu_kashmir"
)

# Function to separate district and state
split_district_state <- function(df, column_name, state_list) {
  df <- df |>
  mutate(
    # Identify the state part in each entry
    state = sapply(str_split(!!sym(column_name), "_"), function(parts) {
      state_match <- ""
      for (i in seq_along(parts)) {
        candidate <- paste(parts[i:length(parts)], collapse = "_")
        if (candidate %in% state_list) {
          state_match <- candidate
          break
        }
      }
      state_match
    }),
    # Remove the state part from the district name
    district = sapply(str_split(!!sym(column_name), "_"), function(parts) {
      state_match <- ""
      for (i in seq_along(parts)) {
        candidate <- paste(parts[i:length(parts)], collapse = "_")
        if (candidate %in% state_list) {
          state_match <- candidate
          break
        }
      }
      # Keep only the parts before the state name
      district_parts <- parts[1:(which(parts == unlist(str_split(state_match, "_"))[1])-1)]
      paste(district_parts, collapse = "_")
    })
  )
  return(df)
}

```

```

}

# Apply the function
india_district_sf <-
  split_district_state(india_district_sf, "district", state_union_names)

## Warning: There were 13 warnings in `stopifnot()`.
## The first warning was:
## i In argument: `district = supply(...)`.:
## Caused by warning in `1:(which(parts == unlist(str_split(state_match, "_"))[1]) - 1)`.:
## ! numerical expression has 2 elements: only the first used
## i Run `dplyr::last_dplyr_warnings()` to see the 12 remaining warnings.

```

Use Python based modules to fuzzy match strings

```

library(reticulate)
library(fuzzywuzzyR)

## If the 'fuzzywuzzyR' package gives the following error: 'attempt to apply non-function' then make su

# Load Python modules
DIFFLIB <- reticulate::import("difflib")
POLYFUZZ <- reticulate::import("polyfuzz")

# Create function that gets the best matches for two vectors of strings, using difflib
get_best_matches <- function(from_vec, to_vec) {

  # Non-vectorized version of the function
  get_best_match <- function(from_string, to_vec) {
    return(DIFFLIB$get_close_matches(from_string, to_vec, 1L, cutoff = 0))
  }

  # Vectorize the function
  unlist(lapply(from_vec, function(from_vec) get_best_match(from_vec, to_vec)))
}

```

Fuzzy Match the state names in spatial and non spatial data set

```

# Get the overall matches for state names that are present in the crime data but not in the spatial data
# difflib_matches_all <- get_best_matches(crime_df$state_name, state_union_names)
difflib_matches_all <- get_best_matches(crime_df$state_name, india_district_sf$state)

# Replace the state_names col the overall matches in the crime dataframe
crime_df$state_name <- difflib_matches_all

```

Fuzzy Match the district names in spatial and non spatial data set

```

difflib_matches_dis <- get_best_matches(crime_df$district, india_district_sf$district)

tfidf_results <- POLYFUZZ$PolyFuzz("TF-IDF")$match(india_district_sf$district, crime_df$district)$get_m

```

Since the Fuzzy Match results for districts are not consistent, manually, match the district names in spatial and non spatial

```
# Check the district names that are present in the crime data but not in the spatial data
(crime_df$district |> unique()) %in% crime_df$district
```

```
## [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [16] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [31] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [46] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [61] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [76] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [91] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [106] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [121] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [136] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [151] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [166] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [181] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [196] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [211] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [226] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [241] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [256] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [271] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [286] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [301] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [316] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [331] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [346] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [361] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [376] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [391] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [406] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [421] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [436] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [451] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [466] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [481] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [496] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [511] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [526] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [541] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [556] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [571] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [586] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [601] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [616] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

```
(crime_df$district |> unique()) [!(crime_df$district |> unique()) %in% india_district_sf$district]
```

```
## [1] "leh (ladakh)" "lahul & spiti"
## [3] "gurdaspur" "shahid bhagat singh nagar"
## [5] "fatehgarh sahib" "tarn taran"
## [7] "sahibzada ajit singh nagar" "tehri garhwal"
## [9] "garhwal" "udham singh nagar"
## [11] "hardwar" "north west"
```

## [13]	"north east"	"new delhi"
## [15]	"south west"	"sawai madhopur"
## [17]	"jyotiba phule nagar"	"gautam buddha nagar"
## [19]	"mahamaya nagar"	"rae bareli"
## [21]	"kanpur dehat"	"kanpur nagar"
## [23]	"allahabad"	"bara banki"
## [25]	"ambedkar nagar"	"shrawasti"
## [27]	"siddharth nagar"	"sant kabir nagar"
## [29]	"sant ravi das nagar(bhadohi)"	"kansiram nagar"
## [31]	"pashchim champaran"	"purba champaran"
## [33]	"saran (chhapra)"	"kaimur (bhabua)"
## [35]	"west kameng"	"east kameng"
## [37]	"papum pare"	"upper subansiri"
## [39]	"west siang"	"east siang"
## [41]	"upper siang"	"lower subansiri"
## [43]	"kurung kumey"	"dibang valley"
## [45]	"lower dibang valley"	"imphal west"
## [47]	"imphal east"	"lawangtlai"
## [49]	"west tripura"	"south tripura"
## [51]	"north tripura"	"west garo hills"
## [53]	"east garo hills"	"south garo hills"
## [55]	"west khasi hills"	"ri bhoi"
## [57]	"east khasi hills"	"jaintia hills"
## [59]	"marigaon"	"karbi anglong"
## [61]	"dima hasao"	"kamrup metropolitan"
## [63]	"darjiling"	"koch bihar"
## [65]	"uttar dinajpur"	"dakshin dinajpur"
## [67]	"bardhaman"	"north 24 parganas"
## [69]	"hugli"	"haora"
## [71]	"south 24 parganas"	"pashchim medinipur"
## [73]	"purba medinipur"	"purbi singhbhum"
## [75]	"pashchimi singhbhum"	"saraikela-kharsawan"
## [77]	"bauda"	"janjgir-champa"
## [79]	"uttar bastar kanker"	"dakshin bastar dantewada"
## [81]	"west nimar"	"narsimhapur"
## [83]	"east nimar"	"banas kantha"
## [85]	"sabar kantha"	"ahmadabad"
## [87]	"panch mahals"	"dohad"
## [89]	"the dangs"	"diu"
## [91]	"dadra & nagar haveli"	"garhchiroli"
## [93]	"mumbai suburban"	"ahmadnagar"
## [95]	"rangareddy"	"mahbubnagar"
## [97]	"warangal"	"east godavari"
## [99]	"west godavari"	"sri potti sriramulu nellore"
## [101]	"y.s.r."	"uttara kannada"
## [103]	"dakshina kannada"	"chamrajnagar"
## [105]	"bangalore rural"	"north goa"
## [107]	"south goa"	"the nilgiris"
## [109]	"nagappattinam"	"virudunagar"
## [111]	"north & middle andaman"	"south andaman"

```

# Clean the district names so that they can be matched in both the data sets
crime_df <- crime_df |>
  mutate(

```

```

district = case_when(
  district == "ahmadabad" ~ "ahmedabad",
  district == "ahmadnagar" ~ "ahmednagar",
  district == "allahabad" ~ "prayagraj",
  district == "ambedkar nagar" ~ "ambedkar_nagar",
  district == "banas kantha" ~ "banaskantha",
  district == "bangalore rural" ~ "bangalore_rural",
  district == "bara banki" ~ "barabanki",
  district == "barddhaman" ~ "barddhaman", # 2017, this was split in purab barddhaman
  district == "bauda" ~ "bharuch",
  district == "chamrajnagar" ~ "chamarajanagar",
  district == "dadra & nagar haveli" ~ "dadra",
  district == "dakshin bastar dantewada" ~ "dantewada",
  district == "dakshin dinajpur" ~ "dakshin_dinajpur",
  district == "dakshina kannada" ~ "dakshina_kannada",
  district == "darjiling" ~ "darjeeling",
  district == "dibang valley" ~ "dibang_valley",
  district == "dima hasao" ~ "dima_hasao",
  district == "dohad" ~ "dahod",
  district == "east garo hills" ~ "east_garo_hills",
  district == "east godavari" ~ "east_godavari",
  district == "east kameng" ~ "east_kameng",
  district == "east khasi hills" ~ "east_khasi_hills",
  district == "east nimar" ~ "khandwa_east_nimar",
  district == "east siang" ~ "east_siang",
  district == "fatehgarh sahib" ~ "fatehgarh_sahib",
  district == "garhchiroli" ~ "gadchiroli",
  district == "garhwal" ~ "pauri_garhwal",
  district == "gautam buddha nagar" ~ "gautam_buddha_nagar",
  district == "gurdaspur" ~ "gurudaspur",
  district == "haora" ~ "howrah",
  district == "hardwar" ~ "haridwar",
  district == "hugli" ~ "hooghly",
  district == "imphal east" ~ "imphal_east",
  district == "imphal west" ~ "imphal_west",
  district == "jaintia hills" ~ "jaintia_hills", # here are two east and west
  district == "janjgir-champa" ~ "janjgir_champa",
  district == "jyotiba phule nagar" ~ "jyotiba_phule_nagar",
  district == "kaimur (bhabua)" ~ "kaimur_bhabua",
  district == "kamrup metropolitan" ~ "kamrup_metropolitan",
  district == "kanpur dehat" ~ "kanpur_dehat",
  district == "kanpur nagar" ~ "kanpur_nagar",
  district == "kansiram nagar" ~ "kanshiram_nagar",
  district == "karbi anglong" ~ "karbi_anglong",
  district == "koch bihar" ~ "koch_bihar",
  district == "kurung kumey" ~ "kurung_kumey",
  district == "lahul & spiti" ~ "lahul_spiti",
  district == "lawangtlai" ~ "lawngtlai",
  district == "leh (ladakh)" ~ "leh",
  district == "lower dibang valley" ~ "lower_dibang_valley",
  district == "lower subansiri" ~ "lower_subansiri",
  district == "mahamaya nagar" ~ "mahamaya_nagar",
  district == "mahbubnagar" ~ "mahabubnagar",

```

```

district == "marigaon" ~ "morigaon",
district == "mumbai suburban" ~ "mumbai_suburban",
district == "nagappattinam" ~ "nagapattinam",
district == "narsinghpur" ~ "narsinghpur",
district == "new delhi" ~ "new_delhi",
district == "north & middle andaman" ~ "north_middle",
district == "north 24 parganas" ~ "north_twenty_four_parganas",
district == "north east" ~ "north_east",
district == "north goa" ~ "north",
district == "north tripura" ~ "north",
district == "north west" ~ "north_west",
district == "panch mahals" ~ "panchmahal",
district == "papum pare" ~ "papum_pare",
district == "pashchim champaran" ~ "pashchim_champaran",
district == "pashchim medinipur" ~ "pashchim_medinipur",
district == "pashchimi singhbhum" ~ "pashchimi_singhbhum",
district == "purba champaran" ~ "purba_champaran",
district == "purba medinipur" ~ "purba_medinipur",
district == "purbi singhbhum" ~ "purbi_singhbhum",
district == "rae bareli" ~ "rae_bareli",
district == "rangareddy" ~ "ranga_reddy",
district == "ri bhoi" ~ "ribhoi",
district == "sabar kantha" ~ "sabarkantha",
district == "sahibzada ajit singh nagar" ~ "sahibzada_ajit_singh_nagar",
district == "sant kabir nagar" ~ "sant_kabeer_nagar",
district == "sant ravi das nagar(bhadohi)" ~ "sant_ravidas_nagar_bhadohi",
district == "saraikela-kharsawan" ~ "saraikela_kharsawan",
district == "saran (chhapra)" ~ "saran",
district == "sawai madhopur" ~ "sawai_madhopur",
district == "shahid bhagat singh nagar" ~ "shahid_bhagat_singh_nagar",
district == "shrawasti" ~ "shravasti",
district == "siddharth nagar" ~ "siddharthnagar",
district == "south 24 parganas" ~ "south_twenty_four_parganas",
district == "south andaman" ~ "south",
district == "south garo hills" ~ "south_garo_hills",
district == "south goa" ~ "south",
district == "south tripura" ~ "south",
district == "south west" ~ "south_west",
district == "sri potti sriramulu nellore" ~ "sri_potti_sriramulu_nellore",
district == "tarn taran" ~ "tarn_taran",
district == "tehri garhwal" ~ "tehri_garhwal",
district == "the dangs" ~ "the_dangs",
district == "the nilgiris" ~ "the_nilgiris",
district == "udham singh nagar" ~ "udam_singh_nagar",
district == "upper siang" ~ "upper_siang",
district == "upper subansiri" ~ "upper_subansiri",
district == "uttar bastar kanker" ~ "uttar_bastar_kanker",
district == "uttar dinajpur" ~ "uttar_dinajpur",
district == "uttara kannada" ~ "uttara_kannada",
district == "virudunagar" ~ "virudhunagar",
district == "warangal" ~ "warangal", # there is warangal_urban to be merged
district == "west garo hills" ~ "west_garo_hills",
district == "west godavari" ~ "west_godavari",

```



```

    district == "west kameng" ~ "west_kameng",
    district == "west khasi hills" ~ "west_khasi_hills",
    district == "west nimar" ~ "khargone_west_nimar",
    district == "west siang" ~ "west_siang",
    district == "west tripura" ~ "west",
    district == "y.s.r." ~ "y_s_r",
    TRUE ~ district
  )
)

```

```

# Check the district names that are present in the crime data but not in the spatial data
(crime_df$district |> unique()) %in% crime_df$district

```

```

## [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [16] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [31] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [46] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [61] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [76] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [91] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [106] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [121] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [136] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [151] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [166] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [181] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [196] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [211] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [226] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [241] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [256] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [271] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [286] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [301] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [316] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [331] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [346] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [361] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [376] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [391] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [406] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [421] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [436] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [451] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [466] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [481] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [496] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [511] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [526] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [541] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [556] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [571] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [586] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [601] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE

```

```
## [616] TRUE TRUE TRUE
(crime_df$district |> unique())[!(crime_df$district |> unique()) %in% india_district_sf$district]

## [1] "jaintia_hills" "barddhaman" "diu" "dadra"
## [5] "warangal"

# Merge the districts in spatial data where the non spatial data refers to only one district (i.e before)
merge_districts <- function(sf_data, districts_to_merge, new_district_name) {
  # Check if all districts to merge exist in the dataset
  if (!all(districts_to_merge %in% sf_data$district)) {
    stop("One or more districts to merge are not present in the dataset.")
  }

  # Merge the geometries of the specified districts
  merged_geom <- sf_data |>
    filter(district %in% districts_to_merge) |>
    summarise(geometry = st_union(geometry), .groups = 'drop')

  # Create a new data frame for the merged district
  merged_district <- sf_data |>
    filter(district %in% districts_to_merge) |>
    select(state) |>
    distinct() |>
    mutate(district = new_district_name)

  # Combine the new merged district with the rest of the data
  result_sf <- rbind(
    sf_data |>
      filter(!district %in% districts_to_merge), # Keep districts that were not merged
    st_sf(merged_district, geometry = merged_geom$geometry) # Add the merged district
  )

  return(result_sf)
}

# Merge Warangal
merged <- merge_districts(india_district_sf, c("warangal_rural", "warangal_urban"), "warangal")

## Warning in data.frame(..., check.names = FALSE): row names were found from a
## short variable and have been discarded
merged <- merge_districts(merged, c("warangal", "warangal"), "warangal")

# Merge Jaintia Hills
merged <- merge_districts(merged, c("east_jaintia_hills", "west_jaintia_hills"), "jaintia_hills")

## Warning in data.frame(..., check.names = FALSE): row names were found from a
## short variable and have been discarded
merged <- merge_districts(merged, c("jaintia_hills", "jaintia_hills"), "jaintia_hills")

# Merge Barddhaman
merged <- merge_districts(merged, c("paschim_barddhaman", "purba_barddhaman"), "barddhaman")

## Warning in data.frame(..., check.names = FALSE): row names were found from a
## short variable and have been discarded
```

```
merged <- merge_districts(merged, c("bardhaman", "bardhaman"), "bardhaman")

glimpse(merged)

## Rows: 723
## Columns: 3
## $ district <chr> "adilabad", "agar_malwa", "agra", "ahmedabad", "ahmednagar", ~
## $ geometry <MULTIPOLYGON [°]> MULTIPOLYGON (((78.84098 19..., MULTIPOLYGON (((~
## $ state <chr> "telangana", "madhya_pradesh", "uttar_pradesh", "gujarat", "m~

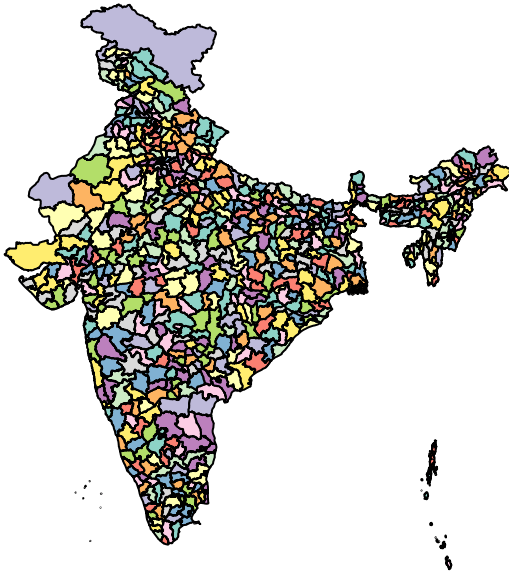
# Set the data to MULTIPOLYGON
merged <- st_cast(merged, "MULTIPOLYGON")

# Print and plot the result
print(merged)

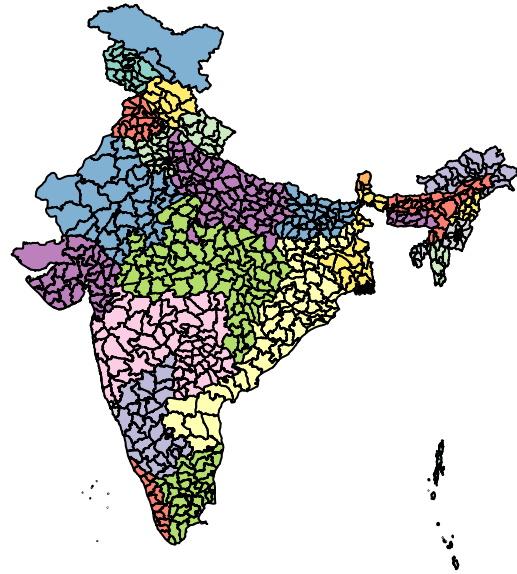
## Simple feature collection with 723 features and 2 fields
## Geometry type: MULTIPOLYGON
## Dimension: XY
## Bounding box: xmin: 68.09382 ymin: 6.755101 xmax: 97.4115 ymax: 37.07719
## Geodetic CRS: WGS 84
## # A tibble: 723 x 3
##   district state geometry
##   <chr> <chr> <MULTIPOLYGON [°]>
## 1 adilabad telangana (((78.84098 19.7619, 78.85224 19.72753, 78.84944 1~
## 2 agar_malwa madhya_pradesh (((76.21246 24.2098, 76.21157 24.19905, 76.21654 2~
## 3 agra uttar_pradesh (((78.19691 27.40256, 78.21589 27.40048, 78.22315 ~
## 4 ahmedabad gujarat (((72.33654 23.10695, 72.35633 23.08818, 72.36506 ~
## 5 ahmednagar maharashtra (((74.67067 19.94678, 74.6719 19.93238, 74.67653 1~
## 6 aizawl mizoram (((93.2045 24.04956, 93.19152 24.01485, 93.16826 2~
## 7 ajmer rajasthan (((74.97485 26.89097, 74.96499 26.87514, 74.95382 ~
## 8 akola maharashtra (((77.61086 20.71787, 77.59075 20.7268, 77.5777 20~
## 9 alappuzha kerala (((76.37292 9.836902, 76.37764 9.832135, 76.38157 ~
## 10 aligarh uttar_pradesh (((78.47747 28.1131, 78.49717 28.10109, 78.51055 2~
## # i 713 more rows

plot(merged)
```

district



state



Merge the spatial and non-spatial data

```
# Merge the non-spatial and spatial data based on the 'district' column
```

```
district_crime_sf <- merged |>  
  st_make_valid() |>  
  group_by(district) |>  
  summarise() |>  
  left_join(crime_df, by = "district")
```

```
# Set the data to MULTIPOLYGON
```

```
district_crime_sf <- st_cast(district_crime_sf, "MULTIPOLYGON")
```

```
# Drop extra col
```

```
district_crime_sf <- district_crime_sf[, -c(2)]
```

```
# Check the structure of the merged data
```

```
district_crime_sf
```

```
## Simple feature collection with 727 features and 9 fields
```

```
## Geometry type: MULTIPOLYGON
```

```
## Dimension: XY
```

```
## Bounding box: xmin: 68.09382 ymin: 6.7551 xmax: 97.4115 ymax: 37.07719
```

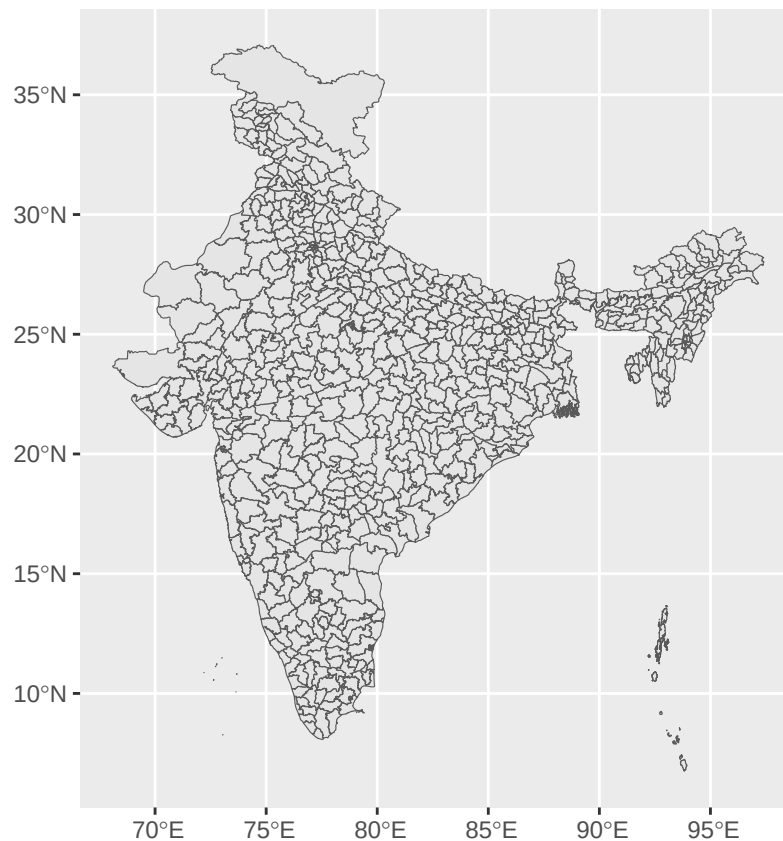
```
## Geodetic CRS: WGS 84
```

```
## # A tibble: 727 x 10
```

```
##   district state_name censuscode rr_2020 ll_2020 ul_2020 rr_2022 ll_2022  
##   <chr>      <chr>      <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>  
## 1 adilabad  andhra_pradesh    532    0.64    0.58    0.69    0.74    0.69
```

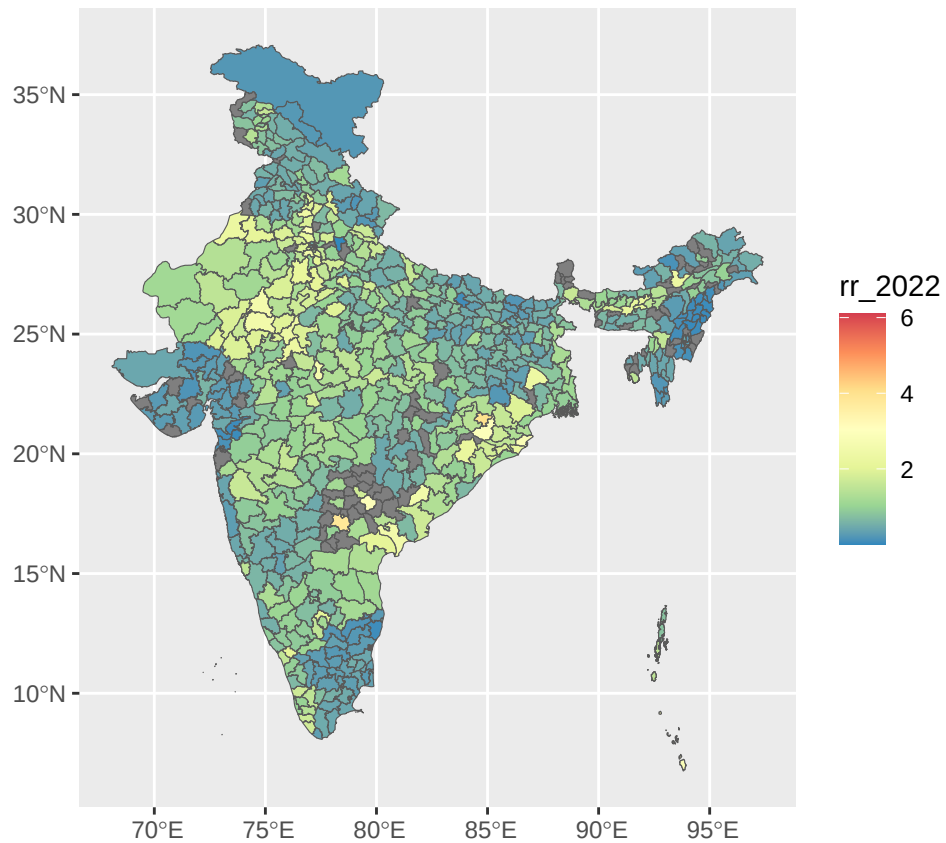
```
## 2 agar_malwa <NA> NA NA NA NA NA
## 3 agra uttar_pradesh 146 0.77 0.72 0.81 1.01 0.97
## 4 ahmedabad gujarat 474 0.77 0.73 0.81 0.62 0.59
## 5 ahmednagar maharashtra 522 1.2 1.15 1.26 1.36 1.3
## 6 aizawl mizoram 283 0.52 0.41 0.65 0.44 0.34
## 7 ajmer rajasthan 119 2.28 2.18 2.39 2.55 2.45
## 8 akola maharashtra 501 1.32 1.22 1.42 1.18 1.1
## 9 alappuzha kerala 598 0.84 0.77 0.91 1.02 0.95
## 10 aligarh uttar_pradesh 143 1.43 1.36 1.5 1.83 1.76
## # i 717 more rows
## # i 2 more variables: ul_2022 <dbl>, geometry <MULTIPOLYGON [°]>
```

```
# Plot the map
district_crime_sf |>
  ggplot() +
  geom_sf()
```

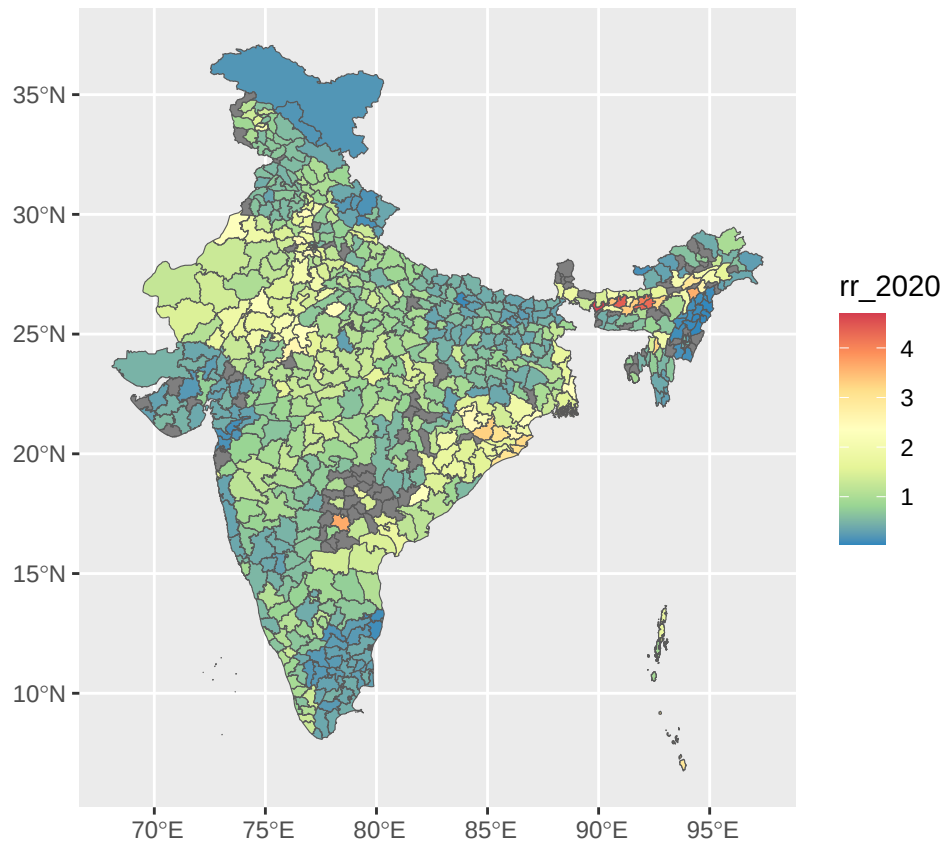


```
# Change color palette for better visualization

# rr_2022 with color palette
district_crime_sf |>
  ggplot() +
  geom_sf(aes(fill = rr_2022)) +
  scale_fill_distiller(palette = "Spectral")
```



```
# rr_2020 with color palette
district_crime_sf |>
  ggplot() +
  geom_sf(aes(fill = rr_2020)) +
  scale_fill_distiller(palette = "Spectral")
```



```
# Plot rr_2022 and rr_2020 side by side using patchwork
```

```
# Create individual plots
```

```
rr_2022 <- district_crime_sf |>
  ggplot() +
  geom_sf(aes(fill = rr_2022)) +
  scale_fill_distiller(palette = "Spectral") +
  labs(title = "Relative Risk 2022")
```

```
rr_2020 <- district_crime_sf |>
  ggplot() +
  geom_sf(aes(fill = rr_2020)) +
  scale_fill_distiller(palette = "Spectral") +
  labs(title = "Relative Risk 2020")
```

```
# add North Arrow and Scale from the ggspatial package
```

```
rr_2022 <- district_crime_sf |>
  ggplot() +
  geom_sf(aes(fill = rr_2022)) +
  scale_fill_distiller(palette = "Spectral") +
  labs(title = "Relative Risk 2022") +
  annotation_north_arrow(location = "tr", which_north = "true", style = north_arrow_minimal) +
  annotation_scale()
```

```
rr_2020 <- district_crime_sf |>
```

```
ggplot() +
  geom_sf(aes(fill = rr_2020)) +
  scale_fill_distiller(palette = "Spectral") +
  labs(title = "Relative Risk 2020") +
  annotation_north_arrow(location = "tr", which_north = "true", style = north_arrow_minimal) +
  annotation_scale()
```

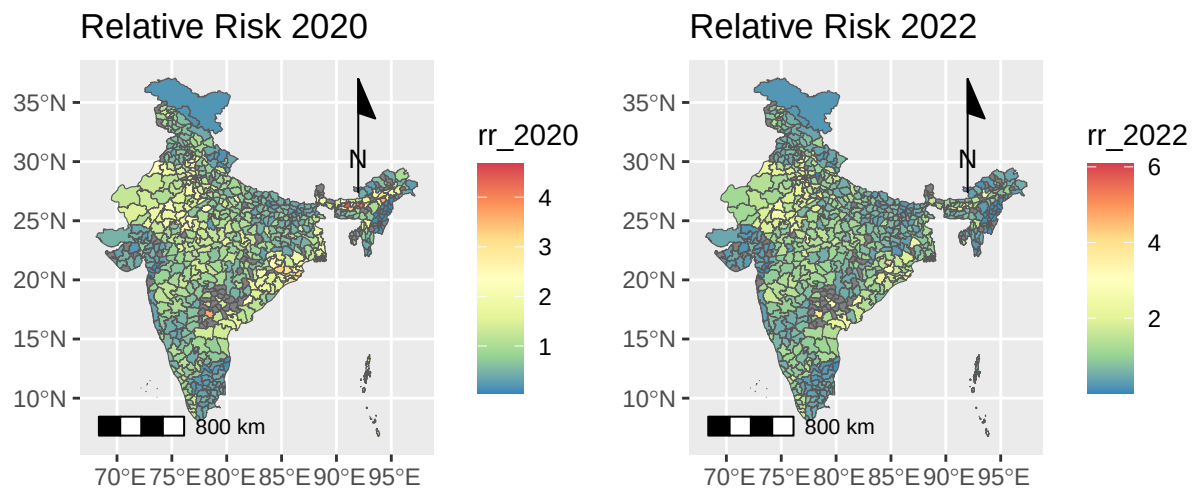
```
# Continuous
```

```
# Combine the plots
```

```
rr_2020 + rr_2022
```

```
## Scale on map varies by more than 10%, scale bar may be inaccurate
```

```
## Scale on map varies by more than 10%, scale bar may be inaccurate
```



```
# Adjust scales for a common legend
```

```
(rr_2020 + rr_2022) +
```

```
  plot_layout(guides = "collect") &
```

```
  scale_fill_distiller(palette = "Spectral", limits = range(c(district_crime_sf$rr_2020, district_crime.
```

```
## Scale for fill is already present.
```

```
## Adding another scale for fill, which will replace the existing scale.
```

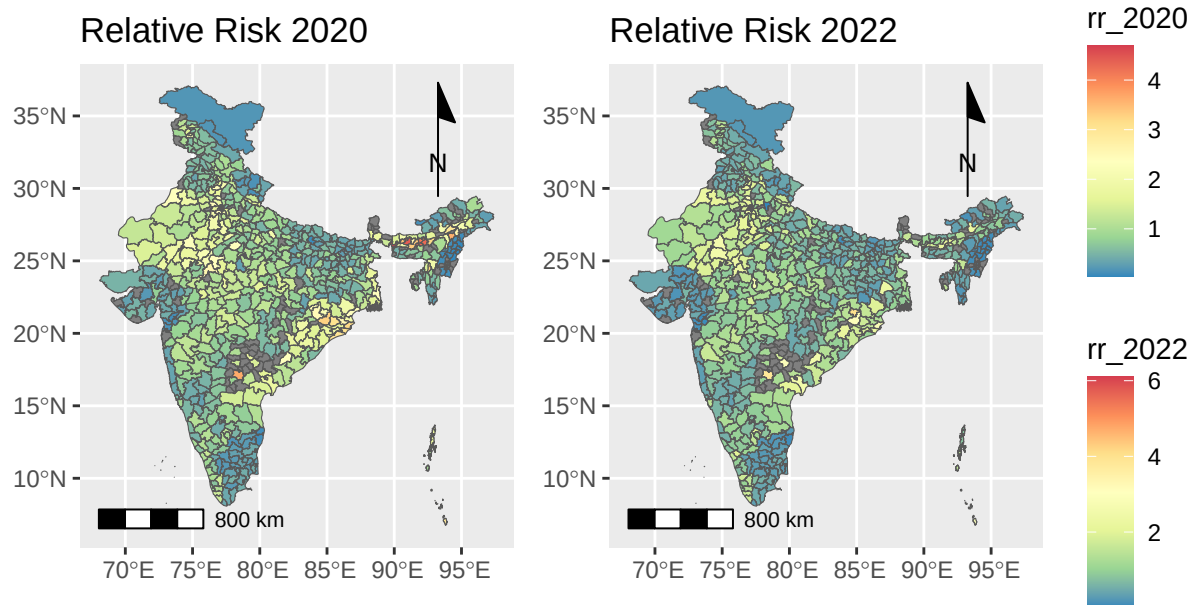
```
## Scale for fill is already present.
```

```
## Adding another scale for fill, which will replace the existing scale.
```

```
## Scale on map varies by more than 10%, scale bar may be inaccurate
```

```
##
```


Scale on map varies by more than 10%, scale bar may be inaccurate



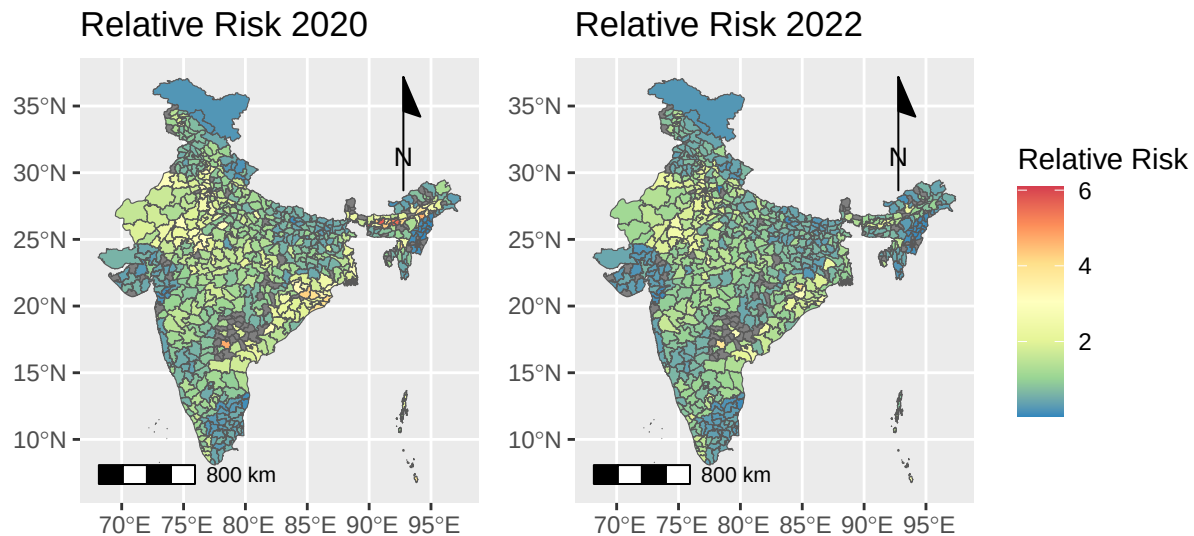
```
# Final plot with title and common legend
rr_crime <- ((rr_2020 + theme(legend.position = "none")) + rr_2022) +
  plot_layout(ncol = 2,
    guides = "collect") +
  plot_annotation(title =
    "Prevalence Distribution of Relative Risk of crimes against women across districts and states in India",
    caption = "Data Source: Data on crimes against women for the period 2020 and 2022 obtained from the National Crime Data Collection",
    theme = theme(plot.title = element_text(hjust = 0.125, size = 12),
      plot.caption = element_text(size = 8))
  ) &
  scale_fill_distiller("Relative Risk", palette = "Spectral", limits = range(c(district_crime_sf$rr_2020, district_crime_sf$rr_2022)))

## Scale for fill is already present.
## Adding another scale for fill, which will replace the existing scale.
## Scale for fill is already present.
## Adding another scale for fill, which will replace the existing scale.

# Display the Continuous plot
rr_crime
```

Scale on map varies by more than 10%, scale bar may be inaccurate
Scale on map varies by more than 10%, scale bar may be inaccurate

evaluation Distribution of Relative Risk of crimes against women across districts and state



: Data on crimes against women for the period 2020 and 2022 obtained from the National Crime Records Bureau (NCRB) of India

```
# Save the plot
ggsave(filename = here("plots", "rr_crime_plot_continuous.png"),
        plot = rr_crime,
        device = "png",
        width = 24,
        height = 9,
        dpi = 300)

## Scale on map varies by more than 10%, scale bar may be inaccurate
## Scale on map varies by more than 10%, scale bar may be inaccurate

# Discrete

# Bin the continuous variables into 5 categories
district_crime_sf$rr_2020_binned <-
  cut(district_crime_sf$rr_2020, breaks = 5, labels = c("Low", "Medium-Low", "Medium", "Medium-High", "High"))

district_crime_sf$rr_2022_binned <-
  cut(district_crime_sf$rr_2022, breaks = 5, labels = c("Low", "Medium-Low", "Medium", "Medium-High", "High"))

# Plot using the binned variables with correct fill aesthetic
plot_2020 <- ggplot(district_crime_sf) +
  geom_sf(aes(fill = rr_2020_binned)) + # Use binned variable for rr_2020
  labs(fill = "Risk Ratio 2020") +
  ggtitle("Relative Risk 2020") +
  annotation_north_arrow(location = "tr", which_north = "true", style = north_arrow_minimal) +
  annotation_scale()
```

```

plot_2022 <- ggplot(district_crime_sf) +
  geom_sf(aes(fill = rr_2022_binned)) + # Use binned variable for rr_2022
  labs(fill = "Risk Ratio 2022") +
  ggtitle("Relative Risk 2022") +
  annotation_north_arrow(location = "tr", which_north = "true", style = north_arrow_minimal) +
  annotation_scale()

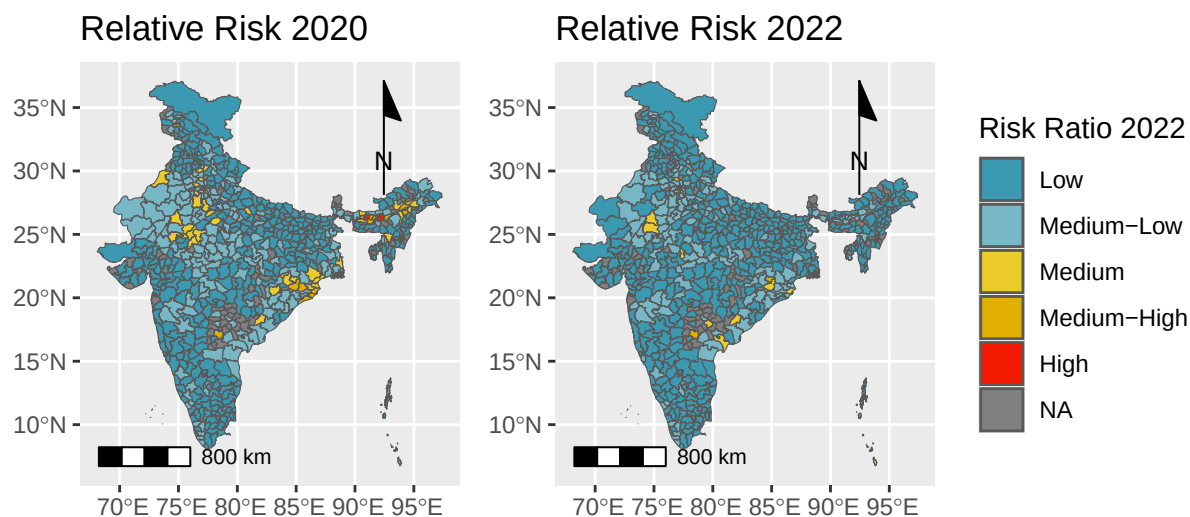
# Combine plots with a shared legend and discrete scale
rr_crime <- ((plot_2020 + theme(legend.position = "none")) + plot_2022) +
  plot_layout(ncol = 2, guides = "collect") +
  plot_annotation(
    title =
      "Prevalence Distribution of Relative Risk of crimes against women across districts and states in India",
    caption = "Data Source: National Crime Records Bureau (NCRB) of India",
    theme = theme_minimal() + theme(
      plot.title = element_text(hjust = 0.125, size = 12), # Center the title
      plot.caption = element_text(size = 8)
    )
  ) & scale_fill_manual(
    values = wes_palette("Zissou1", type = "discrete") # Reverse color direction
  )

# Display the Discrete plot
rr_crime

## Scale on map varies by more than 10%, scale bar may be inaccurate
## Scale on map varies by more than 10%, scale bar may be inaccurate

```

evalence Distribution of Relative Risk of crimes against women across districts and state



Data Source: National Crime Records Bureau (NCRB) of India

```
# Save the plot
ggsave(filename = here("plots", "rr_crime_plot_discrete.png"),
        plot = rr_crime,
        device = "png",
        width = 24,
        height = 9,
        dpi = 300)
```

```
## Scale on map varies by more than 10%, scale bar may be inaccurate
## Scale on map varies by more than 10%, scale bar may be inaccurate
```

Moran's I

```
moran_global <- function(data, column, title = "Scatter Plot of Spatial Lag vs. Original Values") {

  # Check for non-finite values (e.g., NA, NaN, Inf) in the column data
  non_finite_values <- !is.finite(column)

  # Impute missing or non-finite values
  column[!is.finite(column)] <- mean(column, na.rm = TRUE)

  # Replace Inf and -Inf values explicitly
  column[column == Inf] <- max(column[is.finite(column)], na.rm = TRUE)
  column[column == -Inf] <- min(column[is.finite(column)], na.rm = TRUE)

  # Define neighboring polygons
```

```

nb <- poly2nb(data, queen = TRUE)

# Assign weights to the neighbors
lw <- nb2listw(nb, style = "W", zero.policy = TRUE)

# Compute the spatial lag
lag <- lag.listw(lw, column)

# Plot the relationship between the column values and their spatial lags
plot(lag ~ column, pch = 16, asp = 1,
     main = title,                # Use the title parameter
     xlab = "Original Values",    # X-axis label
     ylab = "Spatial Lag",        # Y-axis label
     col = "gray",               # Color of points
     cex = 1.2)                  # Size of points

# Fit a linear regression model
M1 <- lm(lag ~ column)

# Add a regression line to the plot
abline(M1, col = "blue", lwd = 2) # Line color and width

# Output the Moran's I coefficient
moran_i_coefficient <- coef(M1)[2] # Moran's I coefficient
print("The slope of the line is the Moran's I coefficient: ")
print(moran_i_coefficient)

# Add text annotation to the plot for Moran's I value
text(x = max(column) * 0.7,      # Adjust x position
     y = min(lag) + (max(lag) - min(lag)) * 0.1, # Adjust y position
     labels = paste("Moran's I =", round(moran_i_coefficient, 3)), # Display rounded Moran's I value
     col = "red",                # Color of the text
     cex = 1.2)                  # Size of the text

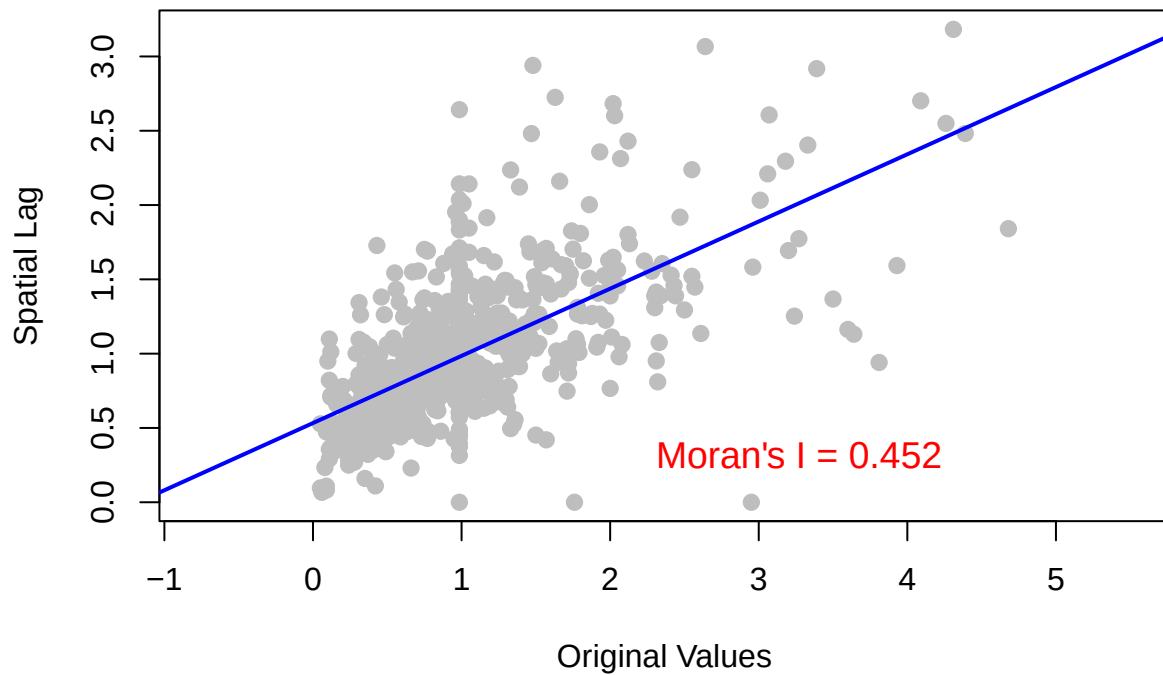
# Moran's I analysis using the analytical method
print(moran.test(column, lw, alternative = "greater"))
}

# Assign the variables
data <- district_crime_sf

# Calculate Global Moran
moran_global_rr_2020 <- moran_global(data, data$rr_2020, title = "Global Moran's I for Relative Risk (2020)")

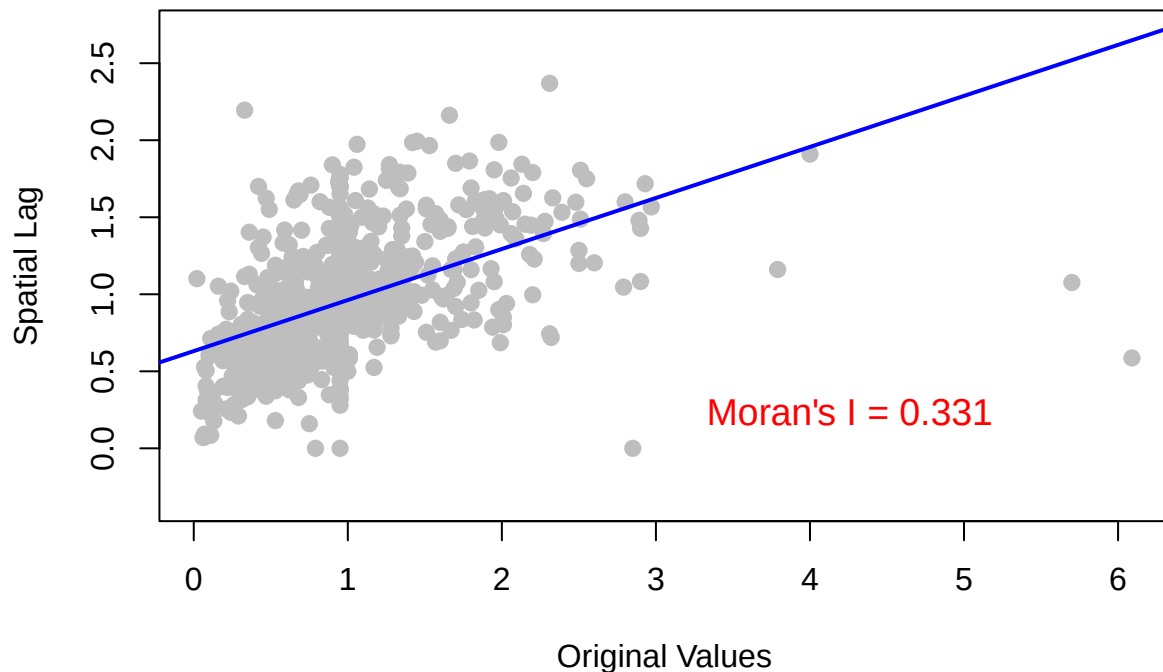
```

Global Moran's I for Relative Risk (2020)



```
## [1] "The slope of the line is the Moran's I coefficient: "
##      column
## 0.4522225
##
## Moran I test under randomisation
##
## data:  column
## weights: lw
## n reduced by no-neighbour observations
##
## Moran I statistic standard deviate = 20.675, p-value < 2.2e-16
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic      Expectation      Variance
##      0.4602269704      -0.0013831259      0.0004984724
moran_global_rr_2022 <- moran_global(data, data$rr_2022, title = "Global Moran's I for Relative Risk (2020)")
```

Global Moran's I for Relative Risk (2022)



```
## [1] "The slope of the line is the Moran's I coefficient: "
##      column
## 0.3312821
##
## Moran I test under randomisation
##
## data:  column
## weights: lw
## n reduced by no-neighbour observations
##
## Moran I statistic standard deviate = 15.229, p-value < 2.2e-16
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic      Expectation      Variance
##      0.3371304826      -0.0013831259      0.0004940854

# Set the filename for saving the plot
filename_2020 <- here("plots", "rr_moran_global_rr_2020.png")
filename_2022 <- here("plots", "rr_moran_global_rr_2022.png")

# Global Moran's I for the Relative Risk 2020
png(filename_2020, width = 800, height = 600) # Open PNG device
moran_global(data, data$rr_2020,
             title = "Global Moran's I for Relative Risk (2020)") # Create the plot

## [1] "The slope of the line is the Moran's I coefficient: "
##      column
```

```
## 0.4522225
##
## Moran I test under randomisation
##
## data: column
## weights: lw
## n reduced by no-neighbour observations
##
## Moran I statistic standard deviate = 20.675, p-value < 2.2e-16
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic      Expectation      Variance
##      0.4602269704      -0.0013831259      0.0004984724
dev.off() # Close the device

## pdf
## 2

# Global Moran's I for the Relative Risk 2022
png(filename_2022, width = 800, height = 600) # Open PNG device
moran_global(data, data$rr_2022,
             title = "Global Moran's I for Relative Risk (2022)") # Create the plot

## [1] "The slope of the line is the Moran's I coefficient: "
##      column
## 0.3312821
##
## Moran I test under randomisation
##
## data: column
## weights: lw
## n reduced by no-neighbour observations
##
## Moran I statistic standard deviate = 15.229, p-value < 2.2e-16
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic      Expectation      Variance
##      0.3371304826      -0.0013831259      0.0004940854
dev.off() # Close the device

## pdf
## 2
```

Local Moran Map

```
moran_local <- function(data, column) {
  # Check for non-finite values (NA, NaN, Inf)
  non_finite_values <- !is.finite(column)

  # Impute with the mean (instance)
  column[!is.finite(column)] <- mean(column, na.rm = TRUE)

  # Replace Inf and -Inf with the maximum and minimum finite values
  column[column == Inf] <-
```



```

max(column[is.finite(column)], na.rm = TRUE)

column[column == -Inf] <-
  min(column[is.finite(column)], na.rm = TRUE)

# Define neighboring polygons # https://mgimond.github.io/simple\_moransI\_example/
nb <- poly2nb(data, queen = TRUE)

# Assign weights to the neighbors
lw <- nb2listw(nb, style = "W", zero.policy = TRUE)

# Compute the (weighted) neighbor mean column values
lag <- lag.listw(lw, column)

# Calculate Local Moran's I statistics for each spatial unit
local_moran <- localmoran(column, lw)

# Add Local Moran's I and p-values to the data set
data$local_moran_I <- local_moran[, 1] # Local Moran's I values
data$local_moran_p <- local_moran[, 5] # p-values

# Classify the Results https://spatialanalysis.github.io/handsonspatialdata/local-spatial-autocorrela

# Define significance level (e.g., 0.05)
significance_level <- 0.05

# Create a classification based on Local Moran's I and p-values
data$local_moran_category <- "Not Significant"

data$local_moran_category[data$local_moran_I > 0 &
  data$local_moran_p <= significance_level] <- "High-High"

data$local_moran_category[data$local_moran_I < 0 &
  data$local_moran_p <= significance_level] <- "Low-Low"

data$local_moran_category[data$local_moran_I > 0 &
  data$local_moran_p > significance_level] <- "High-Low"

data$local_moran_category[data$local_moran_I < 0 &
  data$local_moran_p > significance_level] <- "Low-High"

# Convert to factor for better visualization
data$local_moran_category <-
  factor(
    data$local_moran_category,
    levels = c(
      "Not Significant",
      "High-High",
      "Low-Low",
      "High-Low",
      "Low-High"
    )
  )

```

```

    return(data)
}

# Calculate Local Moran
moran_local_rr_2020 <- moran_local(data, data$rr_2020)
moran_local_rr_2022 <- moran_local(data, data$rr_2022)

# LISA plot 2020
lisa_map_rr_2020 <- tm_shape(moran_local_rr_2020) +
  tm_polygons("local_moran_category",
    palette = wes_palette("IsleofDogs1", n = length(unique(moran_local_rr_2020$local_moran_category))),
    title = "LISA Map for Relative Risk 2020",
    style = "cat") +
  tm_borders() +
  tm_layout(legend.outside = TRUE,
    title = "Local Moran's I Clusters") +
  tm_scale_bar(position = c("left", "bottom"), width = "100 km", text.size = 0.5) +
  tm_compass(position = c("right", "top"), type = "arrow", size = 2)

# LISA plot 2022
lisa_map_rr_2022 <- tm_shape(moran_local_rr_2022) +
  tm_polygons("local_moran_category",
    palette = wes_palette("IsleofDogs1", n = length(unique(moran_local_rr_2022$local_moran_category))),
    title = "LISA Map for Relative Risk 2022",
    style = "cat") +
  tm_borders() +
  tm_layout(legend.outside = TRUE,
    title = "Local Moran's I Clusters") +
  tm_scale_bar(position = c("left", "bottom"), width = "100 km", text.size = 0.5) +
  tm_compass(position = c("right", "top"), type = "arrow", size = 2)

# Combine the plots
combined_lisa_map <- tmap_arrange(
  lisa_map_rr_2020,
  lisa_map_rr_2022,
  ncol = 2,
  # Arrange the maps in 2 columns (side by side)
  widths = 24,
  heights = 9,
  sync = FALSE,
  asp = 0,
  outer.margins = 0.02
)

combined_lisa_map

```

```

## Warning: One tm layer group has duplicated layer types, which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).

```

```

## Scale bar width set to 0.25 of the map width

```

```

## Scale bar set for latitude km and will be different at the top and bottom of the map.

```

```

## Some legend labels were too wide. These labels have been resized to 0.53. Increase legend.width (arg

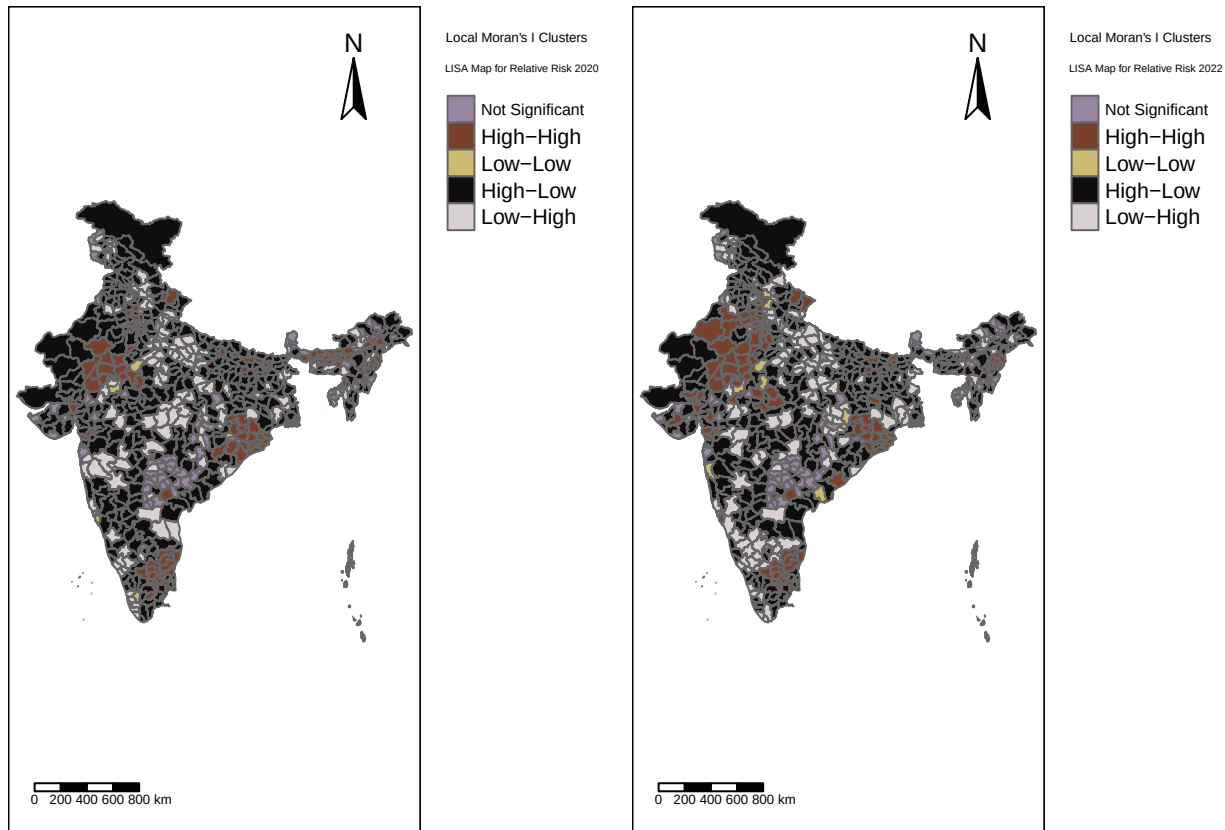
```

```
## Warning: One tm layer group has duplicated layer types, which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).

## Scale bar width set to 0.25 of the map width

## Scale bar set for latitude km and will be different at the top and bottom of the map.

## Some legend labels were too wide. These labels have been resized to 0.53. Increase legend.width (arg
```



```
# Save the plot
tmap_save(combined_lisa_map, filename = here("plots", "rr_combined_lisa_map.png"), scale = 1, width = 24
```

```
## The argument 'units' has been set to "in" since the specified width or height is less than or equal

## Warning: One tm layer group has duplicated layer types, which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).

## Scale bar width set to 0.25 of the map width

## Scale bar set for latitude km and will be different at the top and bottom of the map.

## Warning: One tm layer group has duplicated layer types, which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).

## Scale bar width set to 0.25 of the map width

## Scale bar set for latitude km and will be different at the top and bottom of the map.

## Map saved to D:\R Urban Analytics\SOM 787\Exercices\plots\rr_combined_lisa_map.png
```

```
## Resolution: 7200 by 2700 pixels
```

```
## Size: 24 by 9 inches (300 dpi)
```

Significance Map

```
# Create the Significance Map
```

```
# 2020
```

```
significance_map_rr_2020 <- tm_shape(moran_local_rr_2020) +  
  tm_polygons("local_moran_p",  
    palette = c("#238b45", "#74c476", "#edf8e9"),  
    breaks = c(0, 0.01, 0.05, 1),  
    labels = c("p < 0.01", "0.01 <= p < 0.05", "p >= 0.05"),  
    title = "Significance") +  
  tm_borders() +  
  tm_layout(legend.outside = TRUE,  
    title = "Local Moran Significance Map for Relative Risk 2020") +  
  tm_scale_bar(position = c("left", "bottom"), width = "100 km", text.size = 0.5) +  
  tm_compass(position = c("right", "top"), type = "arrow", size = 2)
```

```
# 2022
```

```
significance_map_rr_2022 <- tm_shape(moran_local_rr_2022) +  
  tm_polygons("local_moran_p",  
    palette = c("#238b45", "#74c476", "#edf8e9"),  
    breaks = c(0, 0.01, 0.05, 1),  
    labels = c("p < 0.01", "0.01 <= p < 0.05", "p >= 0.05"),  
    title = "Significance") +  
  tm_borders() +  
  tm_layout(legend.outside = TRUE,  
    title = "Local Moran Significance Map for Relative Risk 2022") +  
  tm_scale_bar(position = c("left", "bottom"), width = "100 km", text.size = 0.5) +  
  tm_compass(position = c("right", "top"), type = "arrow", size = 2)
```

```
# Combine the plots
```

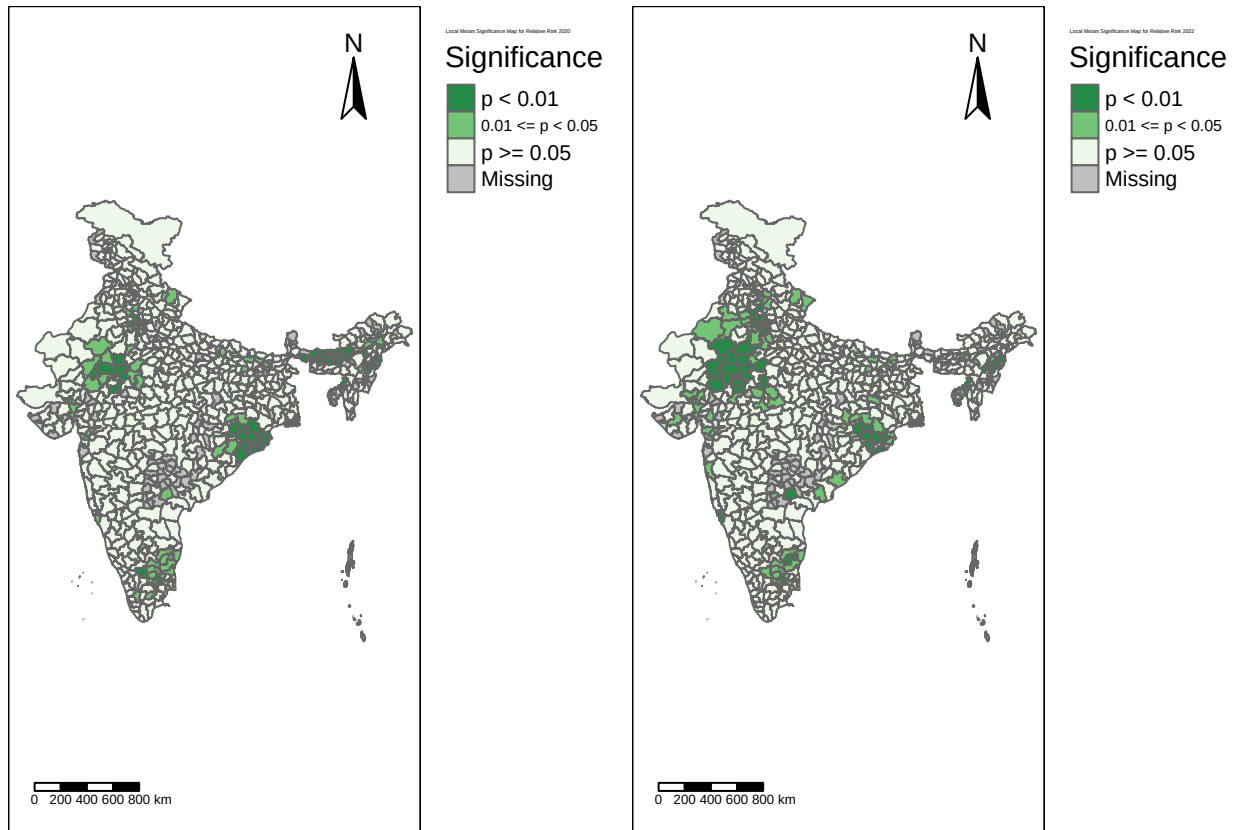
```
combined_significance_map <- tmap_arrange(  
  significance_map_rr_2020,  
  significance_map_rr_2022,  
  ncol = 2,  
  # Arrange the maps in 2 columns (side by side)  
  widths = 24,  
  heights = 9,  
  sync = FALSE,  
  asp = 0,  
  outer.margins = 0.02  
)
```

```
combined_significance_map
```

```
## Warning: One tm layer group has duplicated layer types, which are omitted. To  
## draw multiple layers of the same type, use multiple layer groups (i.e. specify  
## tm_shape prior to each of them).
```

```
## Scale bar width set to 0.25 of the map width
```

```
## Scale bar set for latitude km and will be different at the top and bottom of the map.
## Some legend labels were too wide. These labels have been resized to 0.47. Increase legend.width (arg
## Warning: One tm layer group has duplicated layer types, which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).
## Scale bar width set to 0.25 of the map width
## Scale bar set for latitude km and will be different at the top and bottom of the map.
## Some legend labels were too wide. These labels have been resized to 0.47. Increase legend.width (arg
```



```
# Save the plot
tmap_save(combined_significance_map, filename = here("plots", "rr_combined_significance_map.png"), scale

## The argument 'units' has been set to "in" since the specified width or height is less than or equal
## Warning: One tm layer group has duplicated layer types, which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).
## Scale bar width set to 0.25 of the map width
## Scale bar set for latitude km and will be different at the top and bottom of the map.
## Warning: One tm layer group has duplicated layer types, which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).
## Scale bar width set to 0.25 of the map width
```

```
## Scale bar set for latitude km and will be different at the top and bottom of the map.  
## Map saved to D:\R Urban Analytics\SOM 787\Excercises\plots\rr_combined_significance_map.png  
## Resolution: 7200 by 2700 pixels  
## Size: 24 by 9 inches (300 dpi)
```